

Footstep Detection for Film Sound Production

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ABSTRACT

In this paper, we presented a footstep detection method, which could assist sound editors in positioning the character's footsteps on the timeline of the film. Based on it, a footstep detection system was designed for film sound production. Considering the characteristics of human motion and the needs of film sound production, our method included two parts: data preprocessing and footstep detection modeling. Experiments on various types of shots showed the good generalization and high accuracy of our method. The application evaluation demonstrated the high efficiency of the system.

CCS CONCEPTS

• **Computer systems organization** → **Embedded systems**; *Redundancy*; Robotics; • **Networks** → Network reliability.

KEYWORDS

Footstep Detection, Film Production, Computer Vision

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1 INTRODUCTION

In film production, sound effects play a crucial role in both telling the story and enhancing the immersion of the audience. When shaping characters, their emotions and attitudes could be conveyed by

the characters' sound effects. Among them, the sound of footsteps is inconspicuous but indispensable. Sound effects of footsteps are usually manually produced by professional sound editors nowadays. They use their experience to determine the motion of characters and manually mark each footstep timestamp. Obtaining the correct timestamps of the character's footsteps in the film takes sound editors a lot of time. In addition, for shots with medium, close-up, or extreme close-up views, sound editors are required to be more professional and experienced.

There is limited research in the field of footstep detection, but there have been some research achievements in related domains. In behavior recognition, MMAAction2 [Contributors 2020b] provided a video understanding toolkit that supported various action recognition algorithms, including I3D [Carreira and Zisserman 2017], TSM [Lin et al. 2019], SlowFast [Feichtenhofer et al. 2019], and other action detection works, to estimate the behavior of character's movement. In pose estimation, CMU [Cao et al. 2021] proposed Openpose, which can perform multi-person keypoints detection in real time. Fang [Fang et al. 2017] proposed Alphapose, which can get better results by using the spatial transformation network (STN). In gait estimation, Kordari et al. [Kordari et al. 2015] used inertial sensors for detection, Las et al. [Łaś and Wisniowski 2021] proposed a magnetic field change detection method, Faure et al. [Faure et al. 2001] explored the relationship among realistic animation of synthetic human figures, vivid biomechanical knowledge and physical simulation of task movement. However, in film production, it is unrealistic to use inertial, magnetic or other sensors to collect data from characters.

In this paper, we proposed an image-based footstep detection method and designed a footstep detection system for film sound production as shown in Fig.1. The system was combined with the detection method and the film sound post-production pipeline.

Compared with the previous human footstep detection works, we made the following contributions:

- Proposed a sensorless footstep detection method with high accuracy and good generalization.
- Designed a footstep detection system for footstep sound effect production in film.
- Developed an intuitive GUI for the footstep detection system.

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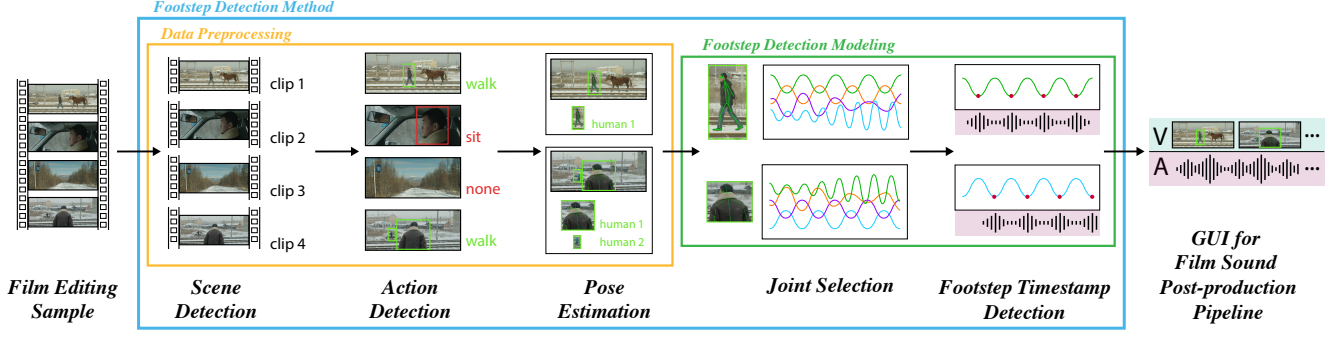


Figure 1: Footstep detection system for film sound production.

2 METHODOLOGY

2.1 Data Preprocessing

The film reference that the sound department gets typically consists of several reels with hundreds of clips. However, not all of these clips need footstep detection as some may not contain characters. We can improve the efficiency of footstep detection by ignoring these unnecessary shots in data preprocessing. Data preprocessing includes scene detection, action detection and pose estimation.

2.1.1 Scene Detection. Divide the film reference into several independent shots using the dynamic threshold method (DTM) [Dimou et al. 2005].

2.1.2 Action Detection. An average 90-minute film typically has over 2,000 shots. It would be time-consuming to perform footstep detection on each individual shot. Therefore, we first use a fast motion detection method to filter out shots that may involve character movements such as walking or running. Then, footstep detection is performed specifically on these selected shots. Here, MMAAction2 [Yue Zhao 2019] is used for action recognition.

2.1.3 Pose Estimation. Estimate the poses of all the characters in each selected shot to obtain their pose data, as shown in Fig.3(a)). In this paper, alphapose [Fang et al. 2022] in the mmpose [Contributors 2020a] framework is used for human pose estimation, for its advantages of efficiency, accuracy, and multi-person support.

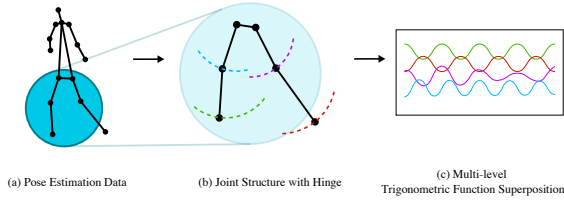


Figure 2: Decomposition of human joints.

2.2 Footstep Detection Modeling

When a person is running or walking, their limbs can be viewed as a simple two-link inverted pendulum model. After having conducted a series of motion capture tests, we found that during stable running and walking, the upper limbs exhibit periodic movements around

the shoulder joint, elbow joint, and wrist joint, while the lower limbs do periodic movements around the hip joint, knee joint, and ankle joint, as shown in Fig. 2. Based on this motion characteristic, we proposed a footstep detection modeling. The model consists of two parts:

- **Joint Selection Function Construction.** Select the joint pose data used for footstep detection. If the set of the selected joints is empty, it means that there are no footsteps in this shot.
- **Footstep Detection.** Detect the footstep timestamp (the time when the footstep appears in the shot) based on the pose data of selected joints.

Multi-person pose data can be obtained by pose estimation. For simplicity, we assume that there is only one character in the selected shot. Applying a single-person footstep detection model to each of them allows for tracking their respective footsteps.

2.2.1 Joint Selection Function construction.

- Filter the pose data of each joint in the frequency domain. Let J_s denote the set of joints on limbs, P^J represent the J joint's pose data ($J \in J_s$). Fast Fourier transform (FFT) is performed on P^J :

$$[f^J, A^J, \phi^J] = \text{FFT}(P^J)$$

where f^J , A^J and ϕ^J are frequencies, amplitudes and phases of the J joint respectively.

Assume f_{max} and f_{min} are constants and represent the maximum and minimum values of joint frequencies that conform to human motion patterns respectively.

$$f_c^J = \{f_{i1}^J, f_{i2}^J, \dots, f_{ik}^J\} \cap [f_{min}, f_{max}],$$

$i_1, i_2, \dots, i_k$ are the subscripts of $A_{i1}^J, A_{i2}^J, \dots, A_{ik}^J$, which are the largest k values of A^J , A_c^J and ϕ_c^J are corresponding amplitudes and phases of f_c^J . Here k is a parameter.

Filtered pose data P_c^J of the J joint (as Fig.3(b)) are computed by equation (1)

$$P_c^J = \text{IFFT}(A_c^J, f_c^J, \phi_c^J) \quad (1)$$

where IFFT is the inverse fast Fourier transform.

- Determining the presence of footsteps in the shot based on filtered joint data and human movement characteristics. If the ratio of amplitudes of the joint J (except for the amplitude

at frequency zero) larger than the threshold A_{Th} is within a given interval, the joint J is selected for footstep detection. Let M_a^J represent the sum of the number of components in A_c^J (except for the amplitude at frequency zero) that are larger than A_{th} ,

$$r_M = \frac{M_a^J(A_c^J > A_{Th})}{L^J}$$

where L^J is the total count of frequencies of J joint. If $r_M \in (0, r_{th}]$, the joint J is selected for footstep detection. Here $r_{th} \in [0, 1]$ is a parameter.

If no joint is selected, there are no footsteps in the shot.

- (c) Joint Selection Function. For the selected joint J in item (b), compute the MSE value E_o^J and SNR value E_{SNR}^J of the joint J 's pose data as follows:

$$E_o^J = \text{MSE}(P^J - P_c^J)$$

$$E_{SNR}^J = 10 * \log_{10} \frac{\|P_c^J\|^2}{\|P_c^J - P^J\|^2}$$

Calculate the maximum similarity S^J of filtered frequency between the joint J and other joints, shown in (2)

$$S^J = \frac{\max_{K \in J_s, K \neq J} [\text{num}(f_c^J = f_c^K)]}{L^J} \quad (2)$$

where f_c^K are the filtered frequencies of the K joint and $\text{num}(f_c^J = f_c^K)$ is the count of components in f_c^J and f_c^K that are equal.

Compute the penalty value PV^J of J joint as follows:

$$PV^J = \alpha \cdot E_o^J + \beta \cdot E_{SNR}^J + \gamma \cdot (1 - S^J)$$

where α, β and γ are parameters.

Select at most two joints, whose penalty values are less than those of the other joints, as shown in Fig.3(c). The parameters $k, A_{Th}, \alpha, \beta, \gamma$ are empirical values and can also be predetermined by training.

2.2.2 Footstep Timestamp Detection. If there are joints that have been selected in item (c), based on the amplitude values of joint motion in the time domain, the timestamps of footsteps are obtained by the following (shown in Fig.3(c-d)):

- (i) If only a single joint is selected, the arguments corresponding to the extreme value of the amplitude in the time domain for that joint are the footsteps' timestamps. That is when t_i is satisfied with the condition as shown in (3), it is the footstep timestamp.

$$(P_c^J(t_{i+1}) - P_c^J(t_i))(P_c^J(t_i) - P_c^J(t_{i-1})) < 0 \quad (3)$$

- (ii) If two joints are selected, the arguments corresponding to the local minima of the amplitude in the time domain for the left and right joints are the footsteps' timestamps. Footstep timestamp t_i must be satisfied with the condition as shown in (4).

$$P_c^J(t_{i+1}) - P_c^J(t_i) > 0 \quad \text{and} \quad P_c^J(t_{i-1}) - P_c^J(t_i) > 0 \quad (4)$$

Condition (3) can also be used to obtain the footsteps' timestamps for each joint, either the union or intersection of them as the footstep timestamp set. However, this approach is more affected by noise.

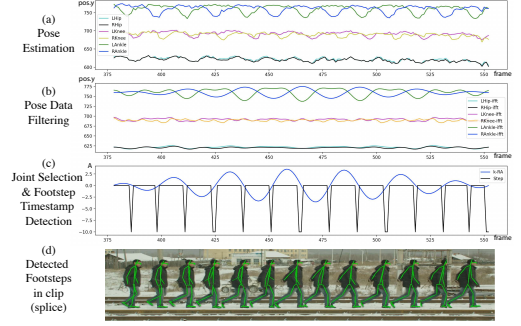


Figure 3: An example to illustrate (a) pose estimation (b) pose data filtering, (c) joint selection and footstep timestamp detection, and (d) the detected footsteps in the clip (splice).

3 FOOTSTEP DETECTION SYSTEM & GUI

We had extensive discussions with the sound production department. In the sound post-production pipeline, the footsteps' timestamps should be represented in the sound production software. Therefore, two sets of footstep timestamp representation schemes are proposed:

- By utilizing a third-party API, markers can be added to the timeline of the shot based on detection results, thus enhancing the efficiency of locating the footstep timestamp.
- utilizing an interactive file format, allowing sound editors to directly map the detection results to a sound library, thereby obtaining the basic footstep soundtrack.

Building upon the detection method and the schemes, we have designed the footstep detection system. Its corresponding GUI interface for sound editors is shown in Fig.4.

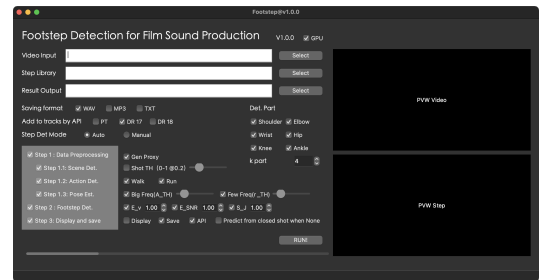


Figure 4: GUI for Footstep Detection System.

4 EVALUATION

4.1 Method Evaluation on Types of Shots

To evaluate the performance of our method, we applied our method to three groups (A, B, C) of test shots, and calculated the accuracy of the detected steps. The indices of accuracy include F_1 score (F_1), the maximum error frame count (Max_F) for the detected footsteps, and the mean ($Mean_F$) of the frame error count compared with their ground truth values. Here, the ground truth frame for footsteps is an interval in practice. Frame error refers to the absolute

Table 1: Experiment Results.

	A (TN = 26)			B (TN = 19)			C (TN = 21)		
	(a)	(b)	(c)	(a)	(b)	(c)	(a)	(b)	(c)
F_1	90.57	90.57	90.57	100.00	100.00	100.00	97.67	93.02	100.00
Max_F	2.00	2.00	2.00	0.00	0.00	0.00	0.00	1.00	0.00
$Mean_F$	1.50	2.00	1.50	0.00	0.00	0.00	0.00	1.00	0.00

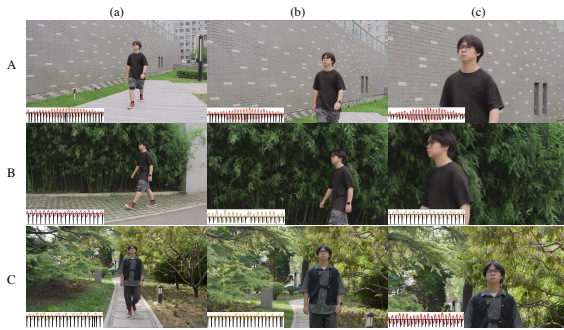
Table 2: Application Evaluation Results.

	D	
	Ours	Manual
Time (min)	3.2	59.7
F_1	95.08	-
Experience	4.67	2.3

value of the difference between the frame of the detected footstep and the closest endpoint of the frame interval.

Each group of test shots consists of three different types of shots. In each group, (a) is a full shot with the character's lower limbs and the ground, (b) is a medium shot without the character's feet, and (c) is a close-up shot with only the character's waist and above. In order to facilitate comparison, (b) and (c) are both cut from (a) in their corresponding group, as shown in Fig.5 (The small picture in the lower left corner of each sub-picture in Fig.5 is the selected joint motion curve (filtered) and the detected footsteps). The characters in different groups face the camera from different angles. The timestamps and the number (TN) of the actual footsteps in the test shots are detected manually in (a) for each group by experienced sound editor.

Table 1 shows that our method can detect the character's footsteps in different types of shots with high accuracy and good generalization. In particular, our method can detect the character's footsteps in shots (c) in which no feet of the characters are in contact with the ground as well as in shots (b) and (c).

**Figure 5: Testing clips and footstep detection results.**

4.2 Application Evaluation

To evaluate the efficiency and ease of use of our footstep detection system in practical applications, we invited three sound editors to detect the footsteps in the test clip manually and with our system respectively, and collected their user interaction experience feedback. Five-point Likert scale is used to evaluate the user's experience with the system, the higher the value, the better the experience.

The test clip (D) consists of parts of shots (b) and (c) from A, B, and C with a duration of 20 seconds. Table 2 (the values of Time and Experience are the average of three sound editors') proves that our system can greatly accelerate the detection of footstep timestamps compared with manual. In addition, user experience evaluations show that our GUI is easy to use and interact with.

5 CONCLUSIONS

In this paper, we proposed an image-based footstep detection method to assist film sound editors in adding footstep sound effects. The application examples proved that the method we proposed can efficiently detect the timestamps of footsteps in different types of shots, saving a lot of time for sound editors in the production of footstep sound effects.

There are some limitations in our method, such as the inaccurate detection of footsteps in dark scene shots, and the inability to detect footsteps without limb joints in the image, for example, in scenes with only close-ups of feet. We will address these issues in our future work.

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